tRNA ms^2io^6A37 Hydroxylation: MiaE, the Terminal O_2-Dependent Diiron Tailoring Step

A review-style synthesis centered on Pseudomonas putida KT2440 MiaE (PP_2188, UniProtKB:Q88KV1; UniPathway UPA00729)

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1. Executive Summary

The anticodon-adjacent position 37 of transfer RNAs that decode UNN codons is one of the most heavily elaborated sites in the tRNA modification landscape. In many bacteria this adenosine is built up in an ordered, three-enzyme relay: **MiaA** adds a dimethylallyl (isopentenyl) group to give N^6-isopentenyladenosine (**i^6A37**); **MiaB**, a radical-SAM methylthiotransferase, installs a 2-methylthio group to give **ms^2i^6A37**; and finally **MiaE** hydroxylates the terminal (C4) carbon of the isopentenyl side chain to give **ms^2io^6A37** (2-methylthio-*cis*-4-hydroxyisopentenyladenosine, historically "2-methylthio-*cis*-ribozeatin") (

P 6362893

P 17910062

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This review concerns only the last of these steps — the **MiaE-catalyzed hydroxylation** that defines UniPathway UPA00729. MiaE is a **non-heme, carboxylate-bridged diiron monooxygenase** with a ferritin-like four-α-helix-bundle catalytic domain. It uses **molecular O_2** and reducing equivalents delivered by a **ferredoxin/ferredoxin-reductase/NAD(P)H** electron-transport chain to perform a highly regio- and stereoselective (>97% *E*) allylic hydroxylation on a substrate that is embedded in a folded tRNA, not a free nucleoside (

P 23906247

P 25453905

P 32785618

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The crystal structure of the in-scope enzyme, *P. putida* MiaE (Pp-MiaE, PP_2188), revealed the diiron site, a dedicated tRNA-recognition linker, and a hydrophobic O_2-delivery tunnel (P 32785618).

Two features make MiaE biologically distinctive. First, because the reaction is **strictly O_2-dependent**, the ms²i⁶A37 ↔ ms²io⁶A37 ratio is a built-in **reporter of aerobiosis and redox state**; in *Salmonella*, hydroxylation occurs aerobically but not anaerobically, and *miaE* mutants cannot grow aerobically on TCA-cycle dicarboxylic acids (P 6362893 P 9620964).

Second, MiaE is a **late, accessory, phylogenetically patchy elaboration**: the i^6A step (MiaA/TRIT1) is conserved across all domains of life, ms^2-thiolation is a bacterial/organellar add-on, and MiaE hydroxylation is confined to a subset of bacteria (e.g., *Salmonella*, *Pseudomonas*) and is absent from *E. coli* and, as far as is known, from eukaryotes (P 17910062 P 21873461 P 24901367).

The strongest evidence base (diiron cofactor, O_2 dependence, stereochemistry, structure, tRNA-context requirement) is solid. The most important open questions concern the **oxygenated diiron intermediate** that abstracts hydrogen, the **native physiological electron donor(s)** in *Pseudomonas*, the **full tRNA substrate set** and recognition determinants, and the **biological meaning** of newly reported side-chain-modified bases and of the aerobiosis-sensing phenotype at the level of specific mRNAs.

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2. Definition and Biological Boundaries

What is included. UPA00729/MiaE comprises exactly one biochemical transformation:

> $ms^2i^6A37\text{-tRNA} + O_2 + 2\,e^- + 2\,H^+ \rightarrow ms^2io^6A37\text{-tRNA} + H_2O$

carried out by a single gene product (PSEPK *miaE* / PP_2188 / Q88KV1) plus the generic redox partners that supply electrons. The substrate is a **fully formed ms^2i^6A37 residue displayed in the anticodon stem-loop (ASL/ACSL)** of specific tRNAs; the product is the hydroxylated ms^2io^6A37 residue. The tailored tRNAs are those that read codons beginning with U (UNN) — classically the isoacceptors for Phe, Tyr, Trp, Ser, Cys and Leu (P 6362893).

What should be treated separately (common conflations).

- **The upstream MiaA and MiaB reactions.** These generate the substrate and are essential *pathway context*, but they are mechanistically unrelated to MiaE (a P-loop dimethylallyltransferase and a radical-SAM/TRAM methylthiotransferase, respectively; P 17910062 P 34526715) and are not part of the single-gene UPA00729 bucket.
- **The eukaryotic i^6A branch and its disease biology.** Human **TRIT1** (a MiaA ortholog) makes i^6A37 on a small subset of cytosolic and mitochondrial tRNAs; its loss causes combined oxidative phosphorylation deficiency 35 (COXPD35) (P 24901367 P 28185376 P 35418828). This is upstream i^6A biology, not MiaE hydroxylation.
- **Cytokinin/plant hormone signaling.** The free base of ms^2io^6A resembles *cis*-zeatin-type cytokinins, and tRNA turnover is one source of *cis*-cytokinins in plants (P 6362893). Cytokinin perception and signaling are a distinct system that merely shares a chemical motif.
- **Other diiron monooxygenases.** MiaE belongs to the broad ferritin-like diiron-carboxylate superfamily that also contains soluble methane/toluene monooxygenases, ribonucleotide reductase R2, Δ9 desaturase and rubrerythrin. These share the four-helix diiron fold but act on entirely different substrates; MiaE is unusual in acting on a macromolecule-embedded target

(32785618). 23906247

Competing definitions. There is essentially no dispute about the reaction itself. Nomenclature varies (the product has been called ms^2io^6A, 2-methylthio-*cis*-hydroxyisopentenyladenosine, or 2-methylthio-*cis*-ribozeatin), and the enzyme has been abbreviated as "MiaE" / "ms^2i^6A37-tRNA monooxygenase" / "(ms^2io^6A37)hydroxylase." Recent mass-spectrometry work has reported additional side-chain-modified species (e.g., a "2-methylthio-methylenethio-N^6-isopentenyladenosine" base; 36640047) and transcriptome-wide mapping of 2-methylthio modifications (36820840); whether any such species are on-pathway MiaE products, alternative oxidation states, or products of other chemistry remains to be established and should be flagged as unsettled.

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3. Mechanistic Overview

Sequence of events (best current model).

1. **Substrate presentation.** MiaE binds the tRNA/ASL and positions the terminal isopentenyl carbon of ms^2i^6A37 near the diiron center. tRNA context is not incidental: the enzyme is ~6,000-fold more efficient (V/K) on an anticodon-stem-loop substrate than on the free i^6A nucleoside (25453905). A dedicated **linker region** identified in Pp-MiaE is important for this recognition (32785618).
2. **O_2 delivery and activation.** O_2 reaches the buried diiron(II) center via a **conserved hydrophobic tunnel** directly visualized by krypton-pressurized crystallography (32785618). Reaction with O_2 generates an oxygenated diiron species (by analogy to other diiron

monooxygenases, a peroxo- and/or high-valent Fe–O intermediate; the exact species in MiaE has not been trapped and is an open question).

3. **C–H activation and hydroxylation.** The activated diiron center abstracts hydrogen from, and installs oxygen at, the **terminal C4 of the isopentenyl side chain**, yielding the allylic alcohol with **>97–98% *E*-stereoselectivity** and no detectable non-specific hydroxylation (P 23906247 P 25453905).
4. **Electron resupply / turnover.** Catalysis requires two electrons per turnover, delivered in vivo by a **ferredoxin (Fd) / ferredoxin-reductase (FdR) / NAD(P)H** chain; in vitro this native chain can be bypassed by a **peroxide shunt** (H_2O_2), which recapitulates the same regio- and stereochemistry, confirming a diiron-oxygen mechanism (P 23906247 P 25453905).

Obligatory vs. conditional vs. accessory. - *Obligatory inputs*: a fully formed **ms^2i^6A37 substrate** (hence obligatory dependence on prior MiaA and MiaB action), **O_2**, and **reducing equivalents**. - *Conditional*: the reaction proceeds **only under aerobic conditions** and is suppressed when iron/cysteine limitation blocks the upstream ms^2-thiolation (leaving i^6A) (P 6362893).
- *Accessory*: the MiaE step itself is **dispensable for viability** — cells lacking MiaE simply accumulate ms^2i^6A37 — so it is a physiological/regulatory tailoring layer rather than a core translational necessity (P 9620964).

4. Major Molecular Players and Active Assemblies

Component	Identity / fold	Role in the step
MiaE (PP_2188, Q88KV1)	Non-heme **carboxylate-bridged diiron** enzyme; **ferritin-like four-α-helix bundle** catalytic domain + tRNA-recognition linker	Binds ms^{2}i^{6}A37-tRNA, activates O_2, performs allylic C4 hydroxylation (P 32785618 P 23906247)
tRNA substrate (ASL of UNN tRNAs)	Folded anticodon stem-loop bearing ms^{2}i^{6}A37	The macromolecular substrate; provides recognition determinants that drive catalytic efficiency (P 25453905)
O_2	Molecular oxygen delivered through a hydrophobic tunnel	Oxygen source and the switch that gates the reaction to aerobiosis (P 32785618 P 6362893)
Ferredoxin / Ferredoxin-reductase / NAD(P)H	Generic [2Fe-2S] Fd + FAD reductase	Electron-transport chain supplying 2 e^- per turnover (P 25453905)
(In vitro) H_2O_2	Peroxide shunt	Bypasses the ETC to drive turnover, diagnostic of diiron-oxygen chemistry (P 23906247)

Upstream players (context, not part of UPA00729): MiaA (dimethylallyl-PP:tRNA dimethylallyltransferase, P-loop/IPPT superfamily) makes i^{6}A37; **MiaB** (radical-SAM methylthiotransferase with a TRAM domain and two [4Fe-4S] clusters) makes ms^{2}i^{6}A37

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P 17910062

P 34526715

P 31899624
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5. Evolutionary and Cell-Biological Variation

A nested, increasingly restricted pathway. The three tailoring layers show a clear conservation gradient:

- **i^6A (MiaA) — ancient and universal.** Isopentenyltransferases are present in bacteria, archaea and eukaryotes; orthologs include yeast Mod5/Tit1, *C. elegans* GRO-1, and human TRIT1 (P 21873461). This is the best representative of the ancestral role of the whole system.
- **ms^2i^6A (MiaB / CDK5RAP1) — bacterial and organellar.** Methylthiolation is added in many bacteria and, in humans, on a subset of **mitochondrial** tRNAs by CDK5RAP1 (a MiaB relative). Notably, *E. coli* stops here (P 17910062).
- **ms^2io^6A (MiaE) — narrowly distributed tip.** Hydroxylation is documented in a subset of bacteria such as *Salmonella* and *Pseudomonas* and is **absent from *E. coli*** and, so far as known, from eukaryotes (P 9620964 P 17910062 P 6362893). MiaE therefore looks like a **late, lineage-specific elaboration** rather than a conserved core function.

Physiological-state variation (the same cell, different conditions). In a single *Salmonella* strain the modification state of A37 tracks the environment: **ms^2io^6A aerobically, ms^2i^6A anaerobically, and only i^6A under iron/cysteine limitation** (P 6362893). Thus the "variation" most relevant to MiaE is not primarily across cell types but across **metabolic/redox states**, because O_2 availability directly gates the enzyme.

Alternative routes. No enzyme is known to reach ms^2io^6A37 by chemistry other than MiaE's diiron hydroxylation; the outcome (an allylic hydroxyl on the side chain) is not known to be produced by an independent route in bacteria. This contrasts with many tRNA modifications that have convergent, non-orthologous solutions.

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6. Constraints, Dependencies, and Failure Modes

Ordering constraints (why the sequence is fixed). - MiaE requires an intact ms^2i^6A37; it **cannot act before MiaA and MiaB**. Genetically and chemically, blocking either upstream step removes the substrate (i^6A-only or A37-only tRNA) and abolishes hydroxylation (P 6362893). - The **peroxide-shunt substrate studies** show MiaE will hydroxylate i^6A and analogs lacking the 2-methylthio group in vitro, but it strongly prefers the tRNA-embedded ms^2i^6A context; the physiologically relevant order is nonetheless MiaA→MiaB→MiaE (P 23906247 P 25453905).

Compartment / condition constraints. - **Strict O_2 requirement**: hydroxylation is mutually exclusive with anaerobiosis (P 6362893). This makes the reaction a de facto aerobic switch and explains why *miaE* loss is phenotypically silent for viability but consequential for aerobic metabolism. - **Substrate specificity**: only tRNAs that carry ms^2i^6A37 (UNN decoders) are candidates; MiaE does not act on bulk RNA or DNA.

Failure modes and their consequences. - ***miaE* loss** → tRNAs frozen at ms^2i^6A37; cells are viable but **fail to grow aerobically on succinate/fumarate/malate**, with normal transport and TCA/respiratory enzymes — i.e., the defect is in translational tuning, not metabolism per se (P 9620964).
- **Upstream loss of i^6A/ms^2i^6A (miaA/miaB)** has broader consequences on decoding: it reduces translational fidelity/efficiency of UNN codons, increases **+1 frameshifting** (synergizing with loss of wobble queuosine), promotes protein aggregation, and lowers virulence in several pathogens by impairing translation of key regulators (e.g., *Shigella* VirF; *Agrobacterium* vir induction)
(P 42391044

P 10692168

P 1735704).
These effects belong to the *upstream* branch but frame why the position-37 modifications matter. - **Regulatory coupling upstream**: MiaA itself is post-transcriptionally controlled (e.g., by the CsrA/CsrB system), showing that flux into the whole pathway is tunable and stress-responsive
(P 40649849).

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7. Controversies and Open Questions

1. **The oxygenated diiron intermediate is not defined.** MiaE is firmly a diiron-oxygen monooxygenase (EPR/UV-vis, peroxide shunt, structure), but the reactive species that performs C–H activation (peroxo vs. high-valent Fe–O) has not been spectroscopically trapped for MiaE specifically. The mechanism is inferred by analogy to better-studied diiron enzymes
(P 23906247

P 32785618).
2. **The native electron donor in *Pseudomonas*.** In vitro turnover uses a generic Fd/FdR/NAD(P)H chain, but the physiological reductase partner(s) and their regulation in *P. putida*

are not established
(P 25453905).

3. **The complete tRNA substrate set and recognition code.** Docking + mutagenesis implicate a linker region and ASL contacts, but a high-resolution MiaE–tRNA complex and a definitive list of substrate isoacceptors across species are lacking
(P 32785618).
4. **Newly reported side-chain species.** Mass-spectrometry has reported additional ms^2-side-chain-modified bases (e.g., a "methylenethio" species;
P 36640047)
and transcriptome-wide ms^2 mapping
(P 36820840).
Whether these are on-pathway MiaE products, oxidation artifacts, or the work of other enzymes is unresolved.
5. **Mechanism of aerobiosis "sensing."** The elegant hypothesis that cells read the hydroxylation status of A37 to gate growth on TCA intermediates is well supported genetically
(P 6362893
P 9620964),
but the downstream molecular chain — which specific mRNAs/attenuators are re-tuned by ms^2io^6A37 vs. ms^2i^6A37 — has not been mapped at modern resolution.
6. **Cross-organism extrapolation caution.** Much biochemistry is from *Salmonella* MiaE while the definitive structure is from *P. putida*; the two are congruent but not identical, and physiology (TCA-growth phenotype) has been characterized mainly in *Salmonella*. Claims should not be generalized to "all bacteria," and eukaryotic i^6A/TRIT1 disease data must not be imported as MiaE biology.

8. Key References

- Buck M, Ames BN. *A modified nucleotide in tRNA as a possible regulator of aerobiosis: synthesis of cis-2-methyl-thioribosylzeatin in the tRNA of Salmonella.* Cell 1984.

  P 6362893

  (Discovery; O_2 dependence; aerobic/anaerobic switching; cytokinin link.)
- Persson BC, et al. *The ms2io6A37 modification of tRNA in Salmonella typhimurium regulates growth on citric acid cycle intermediates.* Mol Microbiol 1998.

  P 9620964

  (miaE gene product = tRNA(ms^2io^6A37)hydroxylase; aerobic dicarboxylate-growth phenotype.)
- Kaminska KH, et al. *Structural bioinformatics analysis of enzymes involved in the biosynthesis pathway of ms^2io^6A37 in tRNA.* Proteins 2008.

  P 17910062

  (Ordered MiaA→MiaB→MiaE pathway; MiaE predicted diiron four-helix fold.)
- Corder AL, et al. *Peroxide-shunt substrate-specificity for the Salmonella O_2-dependent tRNA-modifying monooxygenase (MiaE).* Biochemistry 2013.

  P 23906247

  (Diiron cofactor; peroxide shunt; >97% E-stereoselectivity.)
- Subedi BP, et al. *Steady-state kinetics and spectroscopic characterization of enzyme–tRNA interactions for the non-heme diiron tRNA-monooxygenase MiaE.* Biochemistry 2015.

  P 25453905

  (Fd/FdR/NADPH ETC; ~6,000-fold ASL-context rate enhancement.)
- Carpentier P, et al. *Structural, biochemical and functional analyses of tRNA-monooxygenase MiaE from Pseudomonas putida.* Nucleic Acids Res 2020.

  P 32785618

  (In-scope Pp-MiaE/PP_2188 crystal structure; diiron four-helix bundle; O_2 tunnel; tRNA-recognition linker.)
- Esakova OA, et al. *Structural basis for tRNA methylthiolation by the radical SAM enzyme MiaB.* Nature 2021.

- P 34526715

  (Upstream MiaB mechanism/structure.)
- Zhang B, et al. *First step in catalysis of the radical SAM methylthiotransferase MiaB.* 2020.
- P 31899624

  (Upstream MiaB chemistry.)
- Lamichhane TN, et al. *Plasticity and diversity of tRNA anticodon determinants of substrate recognition by eukaryotic A37 isopentenyltransferases.* RNA 2011.
- P 21873461

  (Deep conservation of the i^6A/MiaA branch; Mod5, Tit1, TRIT1, GRO-1.)
- Yarham JW, et al. *Defective i6A37 modification of mitochondrial and cytosolic tRNAs from pathogenic TRIT1 mutations.* PLoS Genet 2014.
- P 24901367

  (Eukaryotic boundary; TRIT1/COXPD disease.)
- Durand JM, et al. *tRNA modification, temperature and DNA superhelicity target virF in Shigella.* Mol Microbiol 2000.
- P 10692168

  (Position-37 modification and virulence, upstream branch.)
- Sun Y, et al. *Defective queuosine and i6A/ms2i6A modification of tRNATyr cause frameshifting and protein aggregation.* 2026.
- P 42391044

  (Decoding fidelity/proteostasis consequences of A37 undermodification.)
- Libiad M, et al. *Identification of 2-methylthio-methylenethio-N^6-… (a new modified base).* 2023.
- P 36640047

  (Newly reported side-chain species; boundary/uncertainty.)
- Fang L, et al. *Transcriptome-wide mapping of 2-methylthio modifications.* 2023.
- P 36820840

  (ms^2 mapping; boundary/uncertainty.)

- Aubee JI, et al. *Post-transcriptional regulation of MiaA by CsrA and sRNA CsrB.* 2025.

  P 40649849

  (Upstream regulatory tuning of pathway flux.)

Scope note: This synthesis is deliberately restricted to the MiaE hydroxylation step (UPA00729). Upstream MiaA/MiaB reactions, eukaryotic i^{6}A/TRIT1 disease biology, and cytokinin signaling are discussed only to delineate boundaries and are not claimed as members of this module. Where evidence derives from *Salmonella* vs. *Pseudomonas*, or from in vitro surrogate substrates, this is stated explicitly to avoid over-generalization.
